

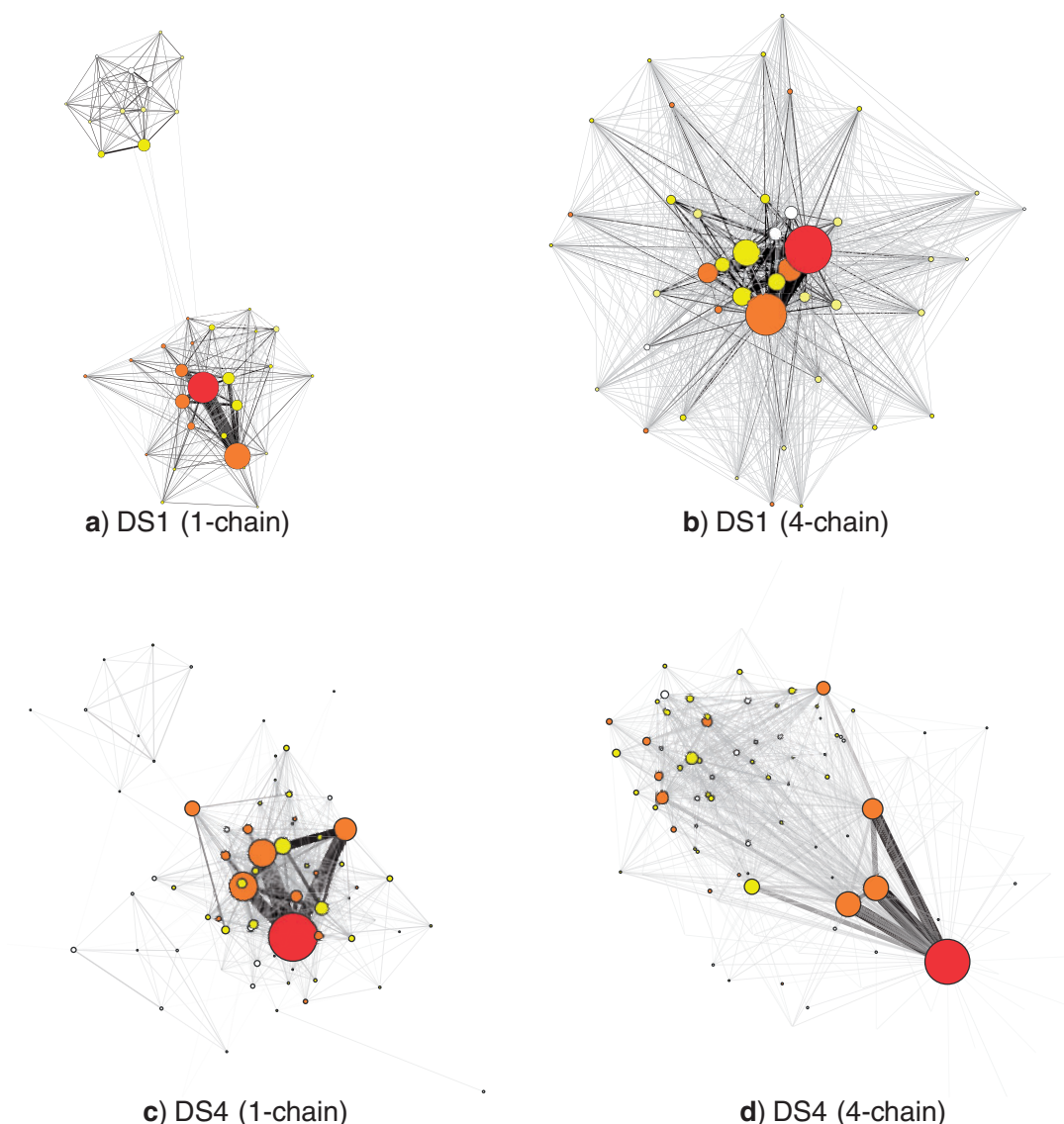


FIGURE 5. Weighted MCMC graphs for DS1 and DS4. Node diameters are scaled relative to posterior probability. Nodes are colored on a red–yellow–white scale (dark–light in the print version) with increasing distance from the topology with highest posterior probability. Edges connect trees in successive 100-iteration samples. Edge thickness and color are proportional to the number of MCMC transitions.

factor (PSRF; see the “Methods” section) for branch lengths was less than 1.01 in all but the 2-run DS2 and DS3 cases and 4 2-run DS7 cases, where the mean PSRF was less than 1.042. Similarly, the ESS for the treelength parameter was greater than 200 except for data sets DS2 and DS3 and 8 of the 4-chain 2-run DS7 cases.

The ASDSF and split frequency error varied considerably over Markov chains of peaky data sets as runs transitioned between peaks. These statistics often dipped below commonly applied thresholds only to increase rapidly after one run began exploring an alternative peak. The subsequent rise and fall of these statistics decreased in magnitude as we gathered a sufficient sample. However, current convergence diagnostics assume that these statistics decrease smoothly and, in particular, do not rise sharply. The first time that an ASDSF cutoff is reached may not

indicate that split frequency estimates are close to their posterior probabilities in peaky posteriors. Moreover, the stopping time for Markov chains is often determined by the first occurrence of a sufficiently small split frequency deviation from golden runs (Höhna and Drummond 2012). This approach may underestimate the time needed to run these chains in the presence of topological peaks because the running observation of the split frequency may get close to the golden run split frequency because of stochasticity.

Metropolis-coupling decreased error when peaky distributions were sampled with a small number of runs. Dataset DS1, in particular, required 7 or more runs to achieve a mean RMSD below 0.02 without Metropolis-coupling but only 3 runs with Metropolis-coupling (Table 4). Much of this error occurred when runs prematurely stopped using a split frequency criterion on